

Slide 1: AI Study Buddy: Document Intelligence

- Learning objective: turn lecture slides into grounded study aids.
- Key pipeline: upload, extract, chunk, retrieve, generate, cite.
- Most testable idea: retrieval quality depends on extraction quality.

Slide 2: Why Students Need Grounded AI

- Students ask local questions about their own notes, not generic web facts.
- A useful study buddy must answer with citations to the uploaded material.
- Risk: fluent answers without evidence create false confidence.

Slide 3: System Architecture

- Frontend sends PDF/TXT/MD uploads to a FastAPI backend.
- Backend stores the file, extracts text, chunks notes, and indexes chunks.
- The AI model receives retrieved context and returns grounded output.

Slide 4: Document Extraction Challenge

- PDF slides can contain text layers, scanned images, tables, code, and figures.
- A single extractor is rarely optimal for every deck.
- Measured text density helps choose between cheap and expensive extraction paths.

Slide 5: Hybrid Extraction Decision

- Path A: pypdf handles normal text PDFs with low cost and low latency.
- Path B: Textract or vision fallback handles low text-density scanned pages.
- Decision rule: fallback when extracted characters per page is below 100.

Slide 6: Table: Extraction Path Trade-Off

- Path | Wins | Loses
- pypdf | pure-text PDFs and Lambda-friendly runtime | weak image/table awareness
- Textract | scanned documents and tables | higher cost per page
- Vision model | figures and diagrams | high latency and token cost

Slide 7: Chunking Strategy

- Current chunking uses slide-aware boundaries when Slide N markers exist.
- Fallback chunking uses sentence boundaries around a target size.
- Chosen default: 700 characters balances context and precision.

Slide 8: Why Chunk Size Matters

- Small chunks improve precision but can lose definitions and examples.
- Large chunks preserve context but can dilute retrieval relevance.
- A good chunk should map to how students ask questions.

Slide 9: Retrieval Quality Metric

- Precision@k measures how many returned citations are relevant.
- StudyBot readiness uses five probe questions and precision@3.
- Target threshold: at least 0.80 on the sample lecture content.

Slide 10: Probe Questions

- Probe 1: What does backpropagation compute?
- Probe 2: What does a confusion matrix compare?
- Probe 3: Why are activation functions used?
- Probe 4: What is overfitting?
- Probe 5: Why does the learning rate matter?

Slide 11: Backpropagation

- Backpropagation computes gradients for each model weight.
- Gradients tell the optimizer how to reduce prediction error.
- This is testable because it connects loss, gradients, and parameter updates.

Slide 12: Gradient Descent

- Gradient descent updates parameters in the opposite direction of the gradient.
- The update size is controlled by the learning rate.
- Too large can diverge; too small can train slowly.

Slide 13: Learning Rate

- The learning rate controls the size of optimizer updates.
- Large values can overshoot minima, while tiny values slow convergence.
- Schedulers reduce the learning rate as training progresses.

Slide 14: Activation Functions

- Activation functions add non-linearity to neural networks.
- Without activations, stacked layers collapse into a linear model.
- Examples include ReLU, sigmoid, tanh, and GELU.

Slide 15: Overfitting

- Overfitting happens when a model memorizes training examples.
- A model that overfits performs poorly on unseen validation data.
- Mitigations include regularization, dropout, data augmentation, and early stopping.

Slide 16: Validation Data

- Validation data estimates generalization before final testing.
- It should not be used to train weights directly.
- Repeated tuning on validation data can leak information.

Slide 17: Confusion Matrix

- A confusion matrix compares predicted labels with actual labels.
- Rows and columns expose true positives, false positives, and false negatives.
- This is useful for imbalanced classification tasks.

Slide 18: Table: Classification Metrics

- Metric | Definition | Interpretation
- Precision | true positives divided by predicted positives | false alarm control
- Recall | true positives divided by actual positives | missed case control
- F1 | harmonic mean of precision and recall | balanced summary

Slide 19: Figure Caption: Training Curves

- Figure 1 shows training loss decreasing across epochs.
- Validation loss decreases at first and rises after epoch six.
- Caption insight: the rise suggests overfitting after epoch six.

Slide 20: Equation: Cross Entropy

- Cross entropy equals the negative sum of target probability times log predicted probability.
- For one-hot labels, only the correct class term remains.
- Lower cross entropy means the model assigns more probability to the correct class.

Slide 21: Code Block: Training Step

- pseudo-code: `loss.backward` computes gradients.
- pseudo-code: `optimizer.step` applies gradients to parameters.
- pseudo-code: `optimizer.zero_grad` clears old gradients before the next batch.

Slide 22: Embeddings

- Embeddings convert text chunks into numeric vectors.
- Similar meanings should produce nearby vectors in embedding space.
- Embedding quality limits retrieval quality.

Slide 23: RAG Flow

- Retrieval augmented generation first searches uploaded chunks.
- The prompt includes retrieved context plus the student question.
- The answer should cite retrieved chunks or slides.

Slide 24: Citation Granularity

- Students expect citations at slide-level granularity.
- Chunk-only citations are acceptable for a prototype but less precise.
- Slide-aware metadata improves trust and debugging.

Slide 25: Prompt Engineering

- Prompts should instruct the model to use only retrieved lecture notes.
- The model should admit when context does not contain the answer.
- JSON prompts help generate structured quizzes and flashcards.

Slide 26: Study Guide Output

- A one-page summary should be concise and exam-oriented.
- The five most testable concepts should include why each is testable.
- The guide should avoid unsupported outside facts.

Slide 27: Flashcards

- Flashcards are best for definitions, contrasts, and short causal explanations.
- Each card should have one clear question and one focused answer.
- Deck quality depends on extracting key concepts from lecture text.

Slide 28: Quiz Generation

- A multiple-choice quiz should include four plausible options.
- The correct option index and explanation must be stored with each question.
- Good distractors test misconceptions instead of random facts.

Slide 29: Dashboard Tracking

- A weekly dashboard tracks which topics the learner studied this week.
- Signals include uploaded documents, asked questions, flashcards, and quizzes.
- Topic extraction from queries gives a lightweight personal study history.

Slide 30: Failure Mode

- Broken query: what is slide 12 about?
- Cause: fixed-size chunks can cut across slide boundaries.
- Mitigation: slide-aware chunking keeps Slide N sections together.

Slide 31: Cost Awareness

- Using Textract for every text PDF overpays for easy documents.
- Using pypdf first saves cost when the text layer is already available.
- Fallback only when text density is low is a defensible hybrid design.

Slide 32: Security and Privacy

- Lecture notes may contain private class material.
- User isolation prevents one student's documents from appearing for another user.
- Production deployments should enforce authentication and least privilege storage access.

Slide 33: Production Vector Store

- The local vector store is useful for demos but does not scale.
- Production can use Bedrock Knowledge Bases or OpenSearch Serverless.
- Metadata filters should include user_id and document_id.

Slide 34: Evaluation Table

- Criterion | Evidence | Current readiness
- Summary | one-page guide plus five concepts | implemented
- Quiz | ten MCQ questions | implemented
- Dashboard | weekly activity | implemented
- Retrieval | precision@3 probes | measured

Slide 35: Demo Script

- Step 1: upload this 40-slide PDF.
- Step 2: generate a study guide and inspect the five testable concepts.
- Step 3: ask about confusion matrices and verify citations.
- Step 4: generate a ten-question quiz and open the dashboard.

Slide 36: Most Testable Concept 1

- Backpropagation computes gradients used by optimizers.
- Why testable: it is a central mechanism in neural network training.

Slide 37: Most Testable Concept 2

- Learning rate controls optimizer update size.
- Why testable: it explains convergence, divergence, and training speed.

Slide 38: Most Testable Concept 3

- Confusion matrices compare predicted labels with actual labels.
- Why testable: it supports precision, recall, and error analysis questions.

Slide 39: Most Testable Concept 4

- Overfitting means memorizing training data and failing to generalize.
- Why testable: it connects validation curves with mitigation strategies.

Slide 40: Most Testable Concept 5

- Hybrid extraction uses text density to choose pypdf or fallback OCR/vision.
- Why testable: it shows architectural judgment and cost-quality trade-offs.